THE FIRST MISSING COTRANSVERSE MAP OCCURS IN DIMENSION FOUR

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Abstract. We correct the proof of Theorem 3.1.16 of *Combinatorics of labelling in higher-dimensional automata*. In the notation of *Towards a theory of natural directed paths*, let $\overline{\square}$ be the category of cubes generated by the coface maps δ_i^α, the symmetry maps σ_i, and the cotransverse degeneracy maps γ_i. The map proposed in the 2010 paper is a composite of two comparators and therefore belongs to $\overline{\square}$. We prove

$$\overline{\square}([3],[3]) = \widehat{\square}_S([3],[3]) \qquad \text{and} \qquad \overline{\square}([4],[4]) \subsetneq \widehat{\square}_S([4],[4]).$$

The equality in dimension three is obtained from the incidence cycle C_6 between the vertices of ranks one and two of the 3-cube. The strict inclusion in dimension four is witnessed by an explicit map constructed from the path graph on four vertices. The Lean formalization is available here: https://github.com/pga0042/Cotransverse-Maps.

Contents

1. Conventions and recalled definitions

We use only the terminology and notation of [3]. Thus $[n] = \{0 < 1\}^n$, the box category $\square$ is generated by the coface maps δ_i^α, and a map $f\colon [m] \to [n]$ of $\mathbf{PoSet}^+$ is *cotransverse* when it takes every directed edge to a directed edge. Equivalently,

$$\overrightarrow{d}_1(x,y) = 1 \quad \Longrightarrow \quad \overrightarrow{d}_1(f(x),f(y)) = 1.$$

The category of all cotransverse maps is denoted by $\widehat{\square}_S$. For $1 \leq i < n$, the symmetry and cotransverse degeneracy maps are

$$\sigma_i(x_1,\dots,x_n) = (x_1,\dots,x_{i-1},x_{i+1},x_i,x_{i+2},\dots,x_n),$$
$$\gamma_i(x_1,\dots,x_n) = (x_1,\dots,x_{i-1},\max(x_i,x_{i+1}),\min(x_i,x_{i+1}),x_{i+2},\dots,x_n).$$

The category generated by the maps δ_i^α, σ_i, and γ_i is denoted by $\overline{\square}$; see [3, Notation 2.15]. In particular, $\overline{\square} \subset \widehat{\square}_S$. Composition is written in the usual functional order: $uv = u \circ v$.

We identify a vertex of $[n]$ with a subset of $\{1,\dots,n\}$. For $A \subset \{1,\dots,n\}$, its characteristic vertex is denoted by χ_A. The rank of χ_A is $|A|$.

This work is AI-assisted.

1.1. **Lemma** (Rank preservation)**.** *Every cotransverse map $f\colon [n] \to [n]$ preserves rank. In particular, it fixes 0_n and 1_n.*

Proof. Choose a maximal chain

$$0_n = x_0 < x_1 < \cdots < x_n = 1_n.$$

Every consecutive pair is a directed edge, so the vertices $f(x_0), \dots, f(x_n)$ form a directed path of length n in $[n]$. Consequently,

$$\mathrm{rk} f(x_n) = \mathrm{rk} f(x_0) + n.$$

Both ranks belong to $\{0, \dots, n\}$, hence $\mathrm{rk} f(x_0) = 0$ and $\mathrm{rk} f(x_n) = n$. Every vertex of rank k lies on a maximal chain and is the k-th vertex of that chain. The same argument therefore gives $\mathrm{rk} f(x) = k$. □

1.2. **Proposition.** *For $0 \le n \le 2$, one has*

$$\overline{\square}([n],[n]) = \widehat{\square}_S([n],[n]).$$

Proof. The cases $n = 0, 1$ are immediate from Lemma 1.1. For $n = 2$, a cotransverse endomorphism fixes 0_2 and 1_2 and is determined by an arbitrary map from the two rank-one vertices to themselves. The four possibilities are realized respectively by

$$\mathrm{id}_{[2]}, \qquad \sigma_1, \qquad \gamma_1, \qquad \sigma_1\gamma_1.$$

□

2. The proposed three-dimensional counterexample

Let $f\colon [3] \to [3]$ be the map used in the proof of [2, Theorem 3.1.16]. Its values, as specified there, are

x	$f(x)$
$(0,0,0)$	$(0,0,0)$
$(1,0,0), (0,1,0), (0,0,1)$	$(0,0,1)$
$(1,1,0), (0,1,1)$	$(0,1,1)$
$(1,0,1)$	$(1,0,1)$
$(1,1,1)$	$(1,1,1)$.

Equivalently,

$$f(x_1,x_2,x_3) = \big(\min(x_1,x_3), \min\big(x_2, \max(x_1,x_3)\big), \max(x_1,x_2,x_3)\big). \tag{1}$$

For $1 \le i < j \le n$, let $c_{ij}\colon [n] \to [n]$ be the comparator which replaces (x_i, x_j) by $(\min(x_i,x_j), \max(x_i,x_j))$ and leaves all other coordinates unchanged. The adjacent comparator is

$$c_{i,i+1} = \sigma_i\gamma_i.$$

Every c_{ij} is therefore in $\overline{\square}$: it is a conjugate of an adjacent comparator by symmetry maps.

2.1. **Proposition.** *The map f of (1) belongs to $\overline{\square}([3],[3])$. More precisely,*

$$f = c_{23}c_{13} = \sigma_2\gamma_2\sigma_2\sigma_1\gamma_1\sigma_2.$$

Proof. The first comparator gives

$$c_{13}(x_1, x_2, x_3) = \big(\min(x_1, x_3), x_2, \max(x_1, x_3)\big).$$

Applying c_{23} yields exactly the right-hand side of (1). Moreover,

$$c_{13} = \sigma_2(\sigma_1\gamma_1)\sigma_2 \quad \text{and} \quad c_{23} = \sigma_2\gamma_2,$$

which proves the displayed word in the generators of $\overline{\square}$. $\square$

The erroneous step in the old proof can now be pinpointed. It obtains the unique factorizations

$$f\delta_1^0 = \delta_1^0(\sigma_1\gamma_1), \qquad f\delta_1^1 = \delta_3^1\sigma_1,$$

and treats the inequality $\sigma_1\gamma_1 \neq \sigma_1$ between the two residual maps $[2] \to [2]$ as an obstruction to expressing f as a composite of the σ_i and γ_i. This implication is false: both displayed identities hold for the generated map of Proposition 2.1. More generally, when a coface map is moved through a word of symmetry and cotransverse degeneracy maps, both its position and the resulting lower-dimensional map may depend on $\alpha \in \{0, 1\}$. Thus the map in the 2010 paper is cotransverse, but it is not a counterexample to equality with $\overline{\square}$.

3. All cotransverse endomorphisms of the three-dimensional cube

We recall the small amount of graph terminology used below. A graph homomorphism is a map on vertices which takes adjacent vertices to adjacent vertices; an endomorphism is a graph homomorphism from a graph to itself. The cycle graph with six vertices is denoted by C_6. These are standard definitions; see, for example, [4, Chapter 1]. Describing an endomorphism of a cycle by the closed walk which is the image of its cyclically ordered vertex set is standard in the study of cycle endomorphism monoids; compare [1, Proposition 1.1 and Proposition 1.9].

Let Γ_3 be the incidence graph whose vertices are the three rank-one vertices and the three rank-two vertices of $[3]$, with an edge for each containment. It is the cycle C_6, and its two parts are the two ranks. By Lemma 1.1, restriction induces a bijection

$$\widehat{\square}_S([3],[3]) \xrightarrow{\cong} \mathrm{End}_{\mathrm{bip}}(C_6), \tag{2}$$

where the right-hand side consists of the endomorphisms preserving the two parts of the bipartition. Indeed, restriction gives such an endomorphism. Conversely, a bipartition-preserving endomorphism of the incidence graph, together with the forced assignments $0_3 \mapsto 0_3$ and $1_3 \mapsto 1_3$, preserves every cover relation and hence defines a cotransverse map.

We now give a structural enumeration and generation of the right-hand side of (2). Number the vertices of C_6 cyclically by $v_0, \ldots, v_5$, with the even vertices in one part. If h preserves the bipartition, then $h(v_0)$ has three possible values. There are signs $\varepsilon_0, \ldots, \varepsilon_5 \in \{-1, 1\}$ such that

$$h(v_{i+1}) = v_{j_i+\varepsilon_i} \quad \text{whenever} \quad h(v_i) = v_{j_i},$$

with indices taken modulo 6. Closure of the walk says

$$\sum_{i=0}^{5} \varepsilon_i \equiv 0 \pmod 6. \tag{3}$$

Thus either all signs are equal, or three signs are 1 and three are -1. It follows at once that

$$\left|\widehat{\square}_S([3],[3])\right| = 3\left(2 + \binom{6}{3}\right) = 66. \tag{4}$$

For completeness, we identify the generators geometrically. The coordinate permutations induce all six automorphisms of C_6 which preserve its bipartition: $\mathrm{Aut}_{\mathrm{bip}}(C_6)$ has order six, and its faithful action on the three rank-one vertices identifies it with the symmetric group $\mathfrak{S}_3$. To check the relevant folding explicitly, take

$$\begin{aligned} v_0 &= \chi_{\{1\}}, & v_1 &= \chi_{\{1,2\}}, & v_2 &= \chi_{\{2\}}, \\ v_3 &= \chi_{\{2,3\}}, & v_4 &= \chi_{\{3\}}, & v_5 &= \chi_{\{1,3\}}. \end{aligned}$$

In this cyclic numbering, the restriction of γ_1 is $(0,1,0,5,4,5)$. If $r(i) = i+2 \pmod 6$, then this tuple is $r^{-1}ar$, where

$$a = (0,1,2,3,2,1), \tag{5}$$

where the tuple records the successive images of $v_0, \ldots, v_5$. The rotation r is induced by the cyclic permutation of the three coordinates. Thus a belongs to the monoid induced by $\overline{\square}([3],[3])$. Three further foldings in the monoid generated by a and the bipartition-preserving automorphisms are

$$b = ara = (2,3,2,1,2,3), \tag{6}$$

$$d = ar^2a = (2,1,0,1,0,1), \tag{7}$$

$$c = (ar^2)b = (0,1,0,1,0,1). \tag{8}$$

3.1. **Lemma.** *Every bipartition-preserving endomorphism of C_6 is generated by $\mathrm{Aut}_{\mathrm{bip}}(C_6)$ and a.*

Proof. The two constant-sign words in (3) give precisely the bipartition-preserving automorphisms. Consider a word with three plus signs and three minus signs, and let

$$S = \{i \in \mathbb{Z}/6 \mid \varepsilon_i = 1\}.$$

For $T \subset \mathbb{Z}/6$ and $t \in \mathbb{Z}/6$, write

$$T + t = \{x + t \mid x \in T\}, \qquad -T-1 = \{-x-1 \mid x \in T\}, \qquad T^c = (\mathbb{Z}/6) \setminus T.$$

Let $\rho(v_i) = v_{i+2}$ and $\tau(v_i) = v_{-i}$, with indices taken modulo 6. Both automorphisms preserve the bipartition of C_6. Precomposition by ρ^{-1} changes the plus-index set according to

$$S \longmapsto S + 2.$$

Postcomposition by τ reverses the orientation of every image edge and therefore gives

$$S \longmapsto S^c.$$

Precomposition by τ sends the oriented edge from v_i to v_{i+1} to the reverse of the edge indexed by $-i-1$. It therefore gives

$$S \longmapsto (-S-1)^c.$$

Combining this operation with postcomposition by τ, we also obtain

$$S \longmapsto -S-1.$$

The action generated by these operations has four orbits on the $\binom{6}{3} = 20$ three-element subsets of $\mathbb{Z}/6$. Abbreviating $\{i,j,k\}$ to ijk, these orbits are

$$\begin{array}{c|cccccc} a & 012 & 234 & 045 & 345 & 015 & 123 \\ b & 034 & 025 & 124 & 125 & 134 & 035 \\ d & 245 & 014 & 023 & 013 & 235 & 145 \\ c & 024 & 135. & & & & \end{array}$$

Indeed, every set displayed in a row is obtained from the first set of that row by repeatedly applying $S \mapsto S+2$ and $S \mapsto S^c$, while $S \mapsto -S-1$ preserves each row. The rows are pairwise disjoint and contain

$$6+6+6+2 = 20 = \binom{6}{3}$$

sets, so no balanced sign word is missing.

The four representatives are realized respectively by the maps a, b, d, c of (5)–(8). By their definitions, these maps belong to the monoid generated by a and $\mathrm{Aut}_{\mathrm{bip}}(C_6)$. Hence every balanced sign word is realized by this monoid. Finally, an even rotation of the target does not change the sign word and realizes each of the three possible values of $h(v_0)$. Consequently every bipartition-preserving endomorphism of C_6 is generated by $\mathrm{Aut}_{\mathrm{bip}}(C_6)$ and a. $\square$

3.2. **Theorem.** *There is no counterexample in dimension three:*

$$\overline{\square}([3],[3]) = \widehat{\square}_S([3],[3]).$$

Both sets have 66 *elements.*

Proof. The inclusion from left to right is part of the definition of $\overline{\square}$. Under (2), the symmetry maps generate $\mathrm{Aut}_{\mathrm{bip}}(C_6)$, and a conjugate of γ_1 restricts to a. Lemma 3.1 therefore shows that every cotransverse endomorphism of $[3]$ belongs to $\overline{\square}$. The cardinality statement is (4). $\square$

3.3. **Corollary.** *For every* $0 \le m \le 3$ *and every* $n \ge m$*, one has*

$$\overline{\square}([m],[n]) = \widehat{\square}_S([m],[n]).$$

Proof. By [3, Proposition 2.5], every cotransverse map $u\colon [m] \to [n]$ factors uniquely as

$$[m] \xrightarrow{v} [m] \xrightarrow{\delta} [n], \qquad \delta \in \square, \quad v \in \widehat{\square}_S([m],[m]).$$

Proposition 1.2 and Theorem 3.2 imply that v belongs to $\overline{\square}$. The coface composite δ belongs to $\square \subset \overline{\square}$, and therefore so does $u = \delta v$. $\square$

4. A cotransverse endomorphism in dimension four

We first record an obstruction satisfied by every non-bijective map in $\overline{\square}([n],[n])$.

4.1. **Definition.** A map $u\colon [n] \to [n]$ *globally folds the pair* $\{p,q\} \subset \{1,\dots,n\}$ if

$$u(\chi_{A\cup\{p\}}) = u(\chi_{A\cup\{q\}}) \quad \text{for every } A \subset \{1,\dots,n\} \setminus \{p,q\}. \tag{9}$$

4.2. **Lemma** (First-fold obstruction)**.** *Every non-bijective map of* $\overline{\square}([n],[n])$ *globally folds at least one pair.*

Proof. A composite from $[n]$ to $[n]$ cannot contain a coface map, since coface maps strictly raise dimension and all maps in a category of cubes have source dimension at most their target dimension. It is therefore a word in symmetry maps and cotransverse degeneracy maps. If it is non-bijective, the word contains a γ_i. Read the word in the order in which its factors are applied, and consider the first γ_i. All earlier factors form a coordinate permutation s. The map γ_i identifies

$$\chi_{B\cup\{i\}} \quad \text{and} \quad \chi_{B\cup\{i+1\}}$$

for every B disjoint from $\{i,i+1\}$. Put $p = s^{-1}(i)$ and $q = s^{-1}(i+1)$. The first part of the word thus identifies the two vertices in (9), and all later factors preserve this equality. Hence the full composite globally folds $\{p,q\}$. $\square$

Let P_4 denote the path graph with vertex set $\{1,2,3,4\}$ and edge set

$$E(P_4) = \big\{\{1,2\},\{2,3\},\{3,4\}\big\}.$$

Define $F\colon [4] \to [4]$, in the subset notation, by

$$F(\chi_A) = \begin{cases} \chi_\varnothing, & |A| = 0,\\ \chi_{\{1\}}, & |A| = 1,\\ \chi_{\{1,2\}}, & |A| = 2 \text{ and } A \in E(P_4),\\ \chi_{\{1,3\}}, & |A| = 2 \text{ and } A \notin E(P_4),\\ \chi_{\{1,2,3\}}, & |A| = 3,\\ \chi_{\{1,2,3,4\}}, & |A| = 4. \end{cases} \tag{10}$$

4.3. **Proposition.** *The map F belongs to* $\widehat{\square}_S([4],[4])$.

Proof. For every cover $A \subset A \cup \{i\}$, the corresponding values in (10) form one of the strict inclusions

$$\varnothing \subset \{1\} \subset \{1,2\} \text{ or } \{1,3\} \subset \{1,2,3\} \subset \{1,2,3,4\}.$$

Thus F is strictly increasing and takes every directed edge to a directed edge. It is cotransverse by definition. $\square$

4.4. **Proposition.** *The map F does not belong to* $\overline{\square}([4],[4])$.

Proof. The map is non-bijective because all four rank-one vertices have the same image. We show that it globally folds no pair, contradicting Lemma 4.2 if it belonged to $\overline{\square}$.

The path P_4 has no pair of twins in the following precise sense: for every $p \neq q$, there is $k \notin \{p,q\}$ which is adjacent to exactly one of p,q. This can be read directly from

$$N(1) = \{2\}, \quad N(2) = \{1,3\}, \quad N(3) = \{2,4\}, \quad N(4) = \{3\}$$

where $N(i)$ is for the open neighbourhood of i, that is, the set of vertices adjacent to i. Choose such a k. Exactly one of $\{p,k\}$ and $\{q,k\}$ is an edge of P_4. Formula (10) then gives

$$F(\chi_{\{p,k\}}) \neq F(\chi_{\{q,k\}}).$$

Taking $A = \{k\}$ in (9) proves that F does not globally fold $\{p,q\}$. Since this holds for every pair, the proof is complete. □

4.5. **Theorem.** *The strict inclusion asserted in [2, Theorem 3.1.16] is true, but its first possible dimension is four:*

$$\overline{\square} \subsetneq \widehat{\square}_S.$$

More precisely,

$$\overline{\square}([3],[3]) = \widehat{\square}_S([3],[3]), \qquad \overline{\square}([4],[4]) \subsetneq \widehat{\square}_S([4],[4]),$$

and the map F of (10) witnesses the second strict inclusion.

Proof. Corollary 3.3 proves that the two categories have the same maps in every hom-set whose source has dimension at most three. Propositions 4.3 and 4.4 give the strict inclusion in dimension four, which implies the strict inclusion of the small categories. □

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